

**Action Bias vs. Utility Theory in Stock Exchange
Market:
A Measure of “Degree”**

by

Quan Zhong

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Dissertation Supervisor: Dunli Li

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Department of Economics

University College London

DECLARATION

I, Quan Zhong, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

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1. ABSTRACT

Background:

Traditional economic theories, particularly expected utility theory, assume that individuals make rational decisions by maximizing expected returns. However, extensive empirical evidence has revealed that real-world investor behavior often deviates from this assumption. Behavioral economics offers alternative explanations, highlighting the role of psychological factors such as cognitive biases and decision-making heuristics. Among these, action bias—the tendency to act even in the absence of clear rational incentives—has drawn increasing attention. This study explores whether investor decisions in the Chinese stock market can be better explained by behavioral models than classical utility-based frameworks.

Method:

Using daily trading data from the Shanghai Stock Exchange Composite Index (SSEC) spanning from December 29, 2014 to December 27, 2024, this research constructs a probit regression model to examine the probability of investor action. Key behavioral variables include the bright line indicator (BL), investor experience (exp), and lagged action (lag_action), while lagged profit (lag_profit) serves as a proxy for marginal return under utility theory. Control variables such as lagged price levels and trading volume are also included. Data pre-processing involves log-transformation, centering, and lag generation, followed by robustness checks such as multicollinearity and residual autocorrelation testing.

Conclusion:

The analysis aims to determine whether action bias provides stronger explanatory power than marginal profit in predicting investor behavior. By contrasting behavioral and traditional economic indicators within the same empirical framework, the study contributes to the understanding of how real-world market decisions are shaped. The findings support a more behaviorally-informed approach to financial modeling, with implications for academic research, investment strategy, and regulatory policy in emerging markets.

Keywords: Behavioral Economics, Probit Regression, Stock Market Decision-Making

2. INTRODUCTION

In classical economic theory, individual decision making is traditionally modeled under the assumption of rationality, where agents are expected to act consistently in their best interests by maximizing expected utility. This framework, often referred to as expected utility theory, has served as a foundational pillar in the field of economics and finance. According to this view, individuals evaluate the potential outcomes of their choices based on objective probabilities and consistent preferences, making decisions aimed at maximizing their utility function. However, mounting empirical evidence over the past several decades has increasingly challenged this assumption, particularly in the context of financial markets, where actual investor behavior often deviates markedly from the predictions of utility-based models.

Behavioral economics has emerged as a powerful alternative framework that seeks to account for these deviations by incorporating insights from psychology and cognitive science into economic modeling. One of the core insights of behavioral economics is that individuals are often influenced by heuristics, biases, and emotional responses that systematically affect their choices. Among the numerous documented behavioral patterns, action bias stands out as a particularly relevant and observable phenomenon in financial decision making. Action bias refers to the human tendency to favor action over inaction, even in situations where inaction may be the optimal choice. In financial markets, this bias manifests itself when investors make trades not based on rational evaluations of profit and loss, but rather due to psychological triggers such as stress, uncertainty, or perceived opportunity.

The purpose of this study is to investigate whether one of the behavioral tendencies: action bias, including such as psychological thresholds (i.e. bright line), experience effects and behavioral inertia can reflect trading behavior in the Chinese stock market more accurately than the classical utility maximization model, and to quantify the extent of the influence of these factors using a probit model.

The broader aim of this research is to contribute to the academic understanding of how behavioral economics complements and sometimes challenges traditional models in explaining financial decision making. As global markets become increasingly complex, it is

essential to recognize the interplay between psychological factors and economic incentives. This study adds to the dialogue by examining action bias and related behavioral constructs in the context of one of the world's largest stock exchange markets.

This study will begin by building on existing research findings and, as such, the next section will review the main theoretical and empirical contributions that form the basis of this study.

3. LITERATURE REVIEW

3.1 Historical Path

Utility Theory Lies in the Field of Rationality

When the question relates to decision making in an economic context, Bernoulli (1954)[4] pioneered the earliest theory by referring to the fact that the means people use in decision making are based on utility. The central idea of utility theory is that an individual, when faced with a choice, rationally assesses the utility (i.e. satisfaction or benefit) that each option may bring and chooses the option that maximizes it. In the course of modern economics, two embodied voices of utility theory, cardinal (Marshall, 1890)[15] and ordinal (Hicks & Allen, 1934)[9], have also emerged, and both have been widely used in (especially) microeconomics decision making. However, utility theory invariably requires an assumption that is both rigorous and more idealized. People are perfectly rational when making decisions; that is, even when they are facing uncertainties, they still make choices based on the probability-weighted expected utility" (Von Neumann & Morgenstern, 1947)[24].

Prospect Theory Suggests Rational Diverts

Although utility theory is widely used, its limitations have also been proposed to be defined, i.e., due to lack of information, cognitive limitations, etc., individuals cannot always make fully rational decisions, instead, they tend to choose the "satisfactory solution" rather than the "optimal solution". This concept is phrased as "bounded rationality" (Simon, 1955)[20]. After this, the concept was applied to behavioral economics and studied in more detail, in which the great work of behavioral economics "Prospect Theory: An Analysis of Decision under Risk" (Kahneman & Tversky, 1988)[12] was published, and in this paper the authors first came and verified the idea that, in the face of risk and uncertainty, people's true actions are often irrational (i.e., they do not follow the expected utility decision rule), and more specifically, people tend to be risk seeking in gains and are risk averse when facing losses.

Irrational Promotes the Idea of Heuristics and Biases

Continuing in concerns under uncertainty, Tversky & Kahneman (1982)[23] suggest that the decision-making process can be complicated, while there are heuristics (mental shortcuts or simplified strategies that people use to make decisions and judgments quickly and efficiently) that can help to foster the decision process. Still, in this "quick" decision-making stage, heuristics may also lead to severe and systematic errors defined as biases. Both heuristics and bias suggest the primordial idea of irrational behavior. (later in the book "Noise: a flaw in human judgment" (Kahneman et al., 2022)[13]), heuristics and biases are included as a book chapter and formally interpreted as a form of decision making under irrationality.

Action Bias is Introduced

In this paper, the topic of research focuses on one of the biases: action bias, and we will tie all of the above together to study the difference in people's decision-making behavior in the stock exchange market under both rationality (utility theory) and irrationality (action bias).

3.2 Action Bias, Model, Relations, and Determinants

Definition of Action Bias

Action bias is defined as a tendency to favor action over inaction, often occurring when people think that they can get benefit or avoid loss from it.

Utility and Action Bias Model

Unlike traditional utility theory, where individuals compare their utility solely based on the value of the final possession, Patt & Zeckhauser (2000)[17] suggest that the occurrence of action bias can only be captured when the "process of gains and losses" are taken into account. To express this more quantitatively, we can set up a model by benchmarking utility theory: in the model, $+(-)I$ refers to positive (negative) impact when no action is taken, $+(-)I_a$ means the positive (negative) impact is due to action. X, Y indicates the presence of two goods (assume both merits), and $\bar{X}, \bar{Y}$ indicates the missing of them, $+(-)X, +(-)Y$ is the process of gain (loss) of X and Y .

Utility decision rule:	Action bias presents when:
$v(+I_a) \perp v(+I)$	$v(+I_a) > v(+I)$
and the goods-valuing system is:	
$v(X, Y) > v(\bar{X}, Y)$, and $v(X, Y) > v(X, \bar{Y})$	$v(+X_a, -Y) > v(\bar{X}, Y)$, and $v(-X, +Y_a) > v(X, \bar{Y})$

Table 3.1: Comparison between utility rule and action bias condition

Introduction to Relations and Determinants

When action is presented, however, questions always arise as: "what if I don't act?" or "Under what circumstances will I act?" At this point, this paper will introduce other biases that stand for inaction and determinants of action bias.

Omission Bias

Action bias occurs when seeking gains or avoiding losses, and omission bias is a natural complement to it (Patt & Zeckhauser, 2000)[17]. The very first founders of this bias, Baron & Ritov (1994)[2], defined it as the perception that loss due to action is worse than inaction, even if the losses are equally severe. This is no doubt an attribute discourages action, but more focuses on the psychological part when loss is already defined. However, this still helps to promote the inaction behavior (Yeung et al., 2021)[25].

Status Quo Bias

The inaction is referred to as status quo bias, which is defined as individuals prefer things to remain the same, or to stick with the current state of affairs, even when changes might lead to better outcomes (Samuelson & Zeckhauser, 1988)[18]. This bias came from the idea of loss aversion (individuals are more sensitive to loss, even when the loss counts the same magnitude as the gains) first mentioned in Prospect Theory (Kahneman & Tversky, 1988)[12]. The higher the loss, the more likely is the loss to trigger the alarm aversion, which, together with the omission bias, promotes the status quo bias. There is certain that this bias is a force against action bias, which we will include in our model

later as a penalty variable.

Action Bias Determinants

Regarding the determinants of action bias, there are three relatively obvious factors summarized from the historical literature: bright line, agency, and potential opportunity for learning.

The bright line is defined as a line that sits clearly between the benefit and the loss; once an individual passes this line, they would gain a certain positive return. In "Action Bias and Environmental Decisions" (Patt & Zeckhauser, 2000)[17], the authors proposed a typical experiment about a choice between whether to clean the river or to preserve air conditions in New England (choose to clean the river will improve the water quality for sure but may in the cost of sacrifice of the current air condition, instead, if choose to preserve air condition, the water quality cannot be improved), and gather individuals choice between these two options, results verified the existence of action bias: individuals favor more about the improvement (round 3/5) rather than the preservation, regardless of the actual benefit that is valued the most. In this sense, the bright line is set to be the definite improvement in air condition, since once across the line, individuals see the actual improvement, while if within the line, the result would be no change (or potential loss) compared to the current situation. Generally speaking, the brighter the line, i.e., the clearer the difference between gains and losses, it is expected to observe a greater tendency to act towards the gains.

Agency is when an individual acts as an agent for others; this idea is observed in "Action bias Among Elite Soccer Goalkeepers: The Case of Penalty Kicks" (Bar-Eli et al., 2007)[3]. In a soccer game context, each player can be seen as an agent of the club/coach, acting under the supervision and scrutiny of the principal; in the article, the authors first collect data on the direction the goalkeeper chooses to pounce on the ball when facing a penalty kick and conclude that in 93. 7% of penalty kicks, the goalkeeper jumps to both sides (which means only 6.3% stays in the center), and the probability of staying in the center to get the best result for the team is 28.7%. This states the irrational behavior, because clearly in round 1/4 of the time, the soccer keeper should stay in the center to get the best result for the team. This shows the presence of an action bias. They then

designed a questionnaire to ask goalkeepers how they would feel to be scored, and most of their answer was 'It feels worse to be scored in the center than on the other sides', and 'If I didn't jump to either side to stop the ball, even if the ball was stopped by me, my team would still blame me for doing nothing'. Therefore, it comes to the conclusion that, under a role as an agent, individuals are more prompted to act in order to show some impact.

The potential opportunity for learning is widely observed in the medical area, where action bias presents as a tendency to treat patience with a complete check of the body (Scherer et al., 2019)[19] or to use the vaccine even when there is a minor signal of having that condition (Gavaruzzi et al., 2010)[7]. Although the bright line (doctors know that the full use of medicine would bring the patient with better condition) and agency (doctors act as agents, actions show their attitudes and caring for patients) both occur in this scenario, a new part also comes to explain the bias for action, and that is the learning opportunity. Doctors tend to treat each patient as an opportunity to learn a new diagnostic case and will often use the full-body examination as a way of understanding the connection between the patient's claimed condition and other possible hidden agendas, and in the case of medication, even if the patient has only the slightest signs, as a proxy for 'seeing where the treatment leads' (Thorpe et al., 2020)[22].

These three determinants form the most important standard measure of the presence of action bias, and in this paper they serve as the main focus of interest (more details are mentioned in research gap).

Notice: In the initial stages of this study, the theoretical framework was based on three key behavioral determinants of action bias: bright line triggers (BL), individual experience (exp), and a sense of agency. Although bright lines and experience are well documented in behavioral economics and have observable counterparts in financial data, agency, defined as the individual's perceived control or autonomy in decision making, proved to be more challenging to quantify within the context of stock market behavior. Due to the lack of a reliable and empirically observable proxy of agency in this setting, it was excluded from the empirical specification.

Instead, as the research progressed, a novel and behaviorally meaningful variable emerged from the data: lagged action (lag_action). This variable captures the inertia of

past behavior and reflects an individual's tendency to repeat previous trading patterns, aligning closely with well-established theories of habit formation and persistence of decisions. Consequently, the final empirical model centers on three core predictors of action: BL, exp, and lag_action. This revised focus not only enhances the empirical tractability of the model, but also strengthens its alignment with observed behavioral dynamics in financial decision making.

3.3 Why Action Bias in the Stock Market

Nature of the Stock Market

This paper focuses on the phenomenon of action bias in the stock exchange market. The stock exchange market is chosen as the setting for this study because the inherent characteristics and unique dynamics of the market make it a fertile ground for observation of action-oriented decision making.

The stock market is with high degree of uncertainty, volatility, and timeliness of decisions (Hsu et al., 2020)[11], this essentially satisfies the background of action bias, and is more close to present the real world situation where there are full of risks and uncertainties. Decisions made under these circumstances will show the two main divergence between traditional utility theory and the biased result.

The numerical figures in stock exchange market also helps to intuitive, as well as allowing for quantitative means to elaborate on the differences between action bias and utility theory.

Most importantly, individuals enter the stock exchange market in search of returns, and before choosing to enter the market, they can observe the performance of the stocks and make a decision through the rate of return, even though the true outcome is uncertain (it may be a loss or a profit), and people's expectations are still induced to be "lucrative". Therefore, in the stock market, we would expect to see a greater degree of action bias than status quo bias, i.e. the significance of action bias in subsequent regression tests.

Overconfidence Bias

Another important reason for choosing the stock market as the background for the

study is that there is another behavioral bias in the stock market: the overconfidence bias (De Bondt & Thaler, 1995)[5]. This idea was first presented in "Judgment under Uncertainty: Heuristics and Biases" (Tversky & Kahneman, 1982)[23], it refers to the tendency of people to overestimate their abilities, knowledge, or the precision of their judgments, and they tend to have too much confidence in their ability to control events.

In some recent studies at the intersection of behavioral economics and financial markets, overconfidence bias has been widely used as a theme in the study of irrational trading in the stock exchange market, where overconfidence bias is reflected in the fact that investors may overestimate their analytical ability, trade stocks frequently or choose risky assets, and think they can beat the market, but actual returns are often below the market average (Parhi & Pal, 2021)[16]. The emergence of this frequent trading phenomenon is also a side effect of the fact that action bias is more significant compared to status quo bias in the stock market.

This paper focuses on action bias rather than overconfidence bias; however, inevitably, there is some overlap and commonality between the two. In this paper, we would like to study the action bias more purely, so we will control for/average individual differences in ability and focus on the determinants of the action bias. However, also the study of overconfidence bias has inspired this post in a recent research paper "The Impact of Investors Overconfidence on Trading Activities and Stock Returns in Pakistan Stock Exchange" (Khan et al., 2021)[14], which uses a proxy-based research methodology on the stock market that we can learn from.

3.4 Research Gap

The Intersection with the Stock Market is New

In summary, the topic of action bias has been well researched and the historical literature provides the known definition of action bias, as well as its determinants. However, there are still very few studies that cross-examine the concept of action bias in the context of the stock exchange market, which is one of the research gaps that this paper will fill.

Empirical Model Constructed, Focus on Degrees

The research question in this paper will not be heavily addressed on the proof of existence of action bias; rather, we will focus more on the term "degree" - the most important point of the research gap that this paper aims to fill. Previous papers conclude to three main determinants of presence of action bias, therefore, we will define the three in stock exchange market using proxy variables in numerical figures, and run through a probit model, to explore how each determinant affects the presence of action bias, and to which extent.

The stock market is a very unique subject, taking the example of soccer goalkeeping, the number of times a goalkeeper needs to keep the ball consecutively may be 3-5 times/each game, however, the stock market is a dynamic market with time series correlation, which also implies that people's actions are coherent in nature (Zhao et al., 2023)[26]. Therefore, through subsequent empirical experiments, we will also propose a new determinant of action bias unique to the stock exchange market that arises as a result of successive decisions/actions: the last-period action (i.e. lagged action).

Compare with the Rational Utility Model

On this basis, we then introduce the utility theory of rational decision making to the stock exchange market to see how the application of utility decision rule affects the tendency towards action and to make comparisons with action bias.

3.5 Conclusion

In conclusion, this paper will be a more in-depth study of action bias and its intersection with the stock exchange market, a direction that none of the previous papers has taken. Based on this context, later in this paper we will find out to what extent the determinants of action bias influence its presence through running regression tests, and most importantly, we will see that action bias causes decisions made to diverge from the predictions of utility theory, and the degree of divergence will be properly defined. Before moving on to the data analysis, first we will discuss the formal theoretical model incorporating the key behavioral factors, in order to set the stage for the subsequent empirical analysis.

4. THEORETICAL MODEL CONSTRUCTION

4.1 Traditional Utility Model (TU):

In classical portfolio theory, an investor's utility combines expected return and risk aversion:

$$U_{TU} = \mathbb{E}[R] - \frac{1}{2}A\sigma^2$$

- ($\mathbb{E}[R]$): Expected return, positive impact on utility.
- (σ^2): Return variance, negative impact scaled by risk aversion coefficient A .

This formulation reflects the rational trade-off: higher expected returns increase utility, while higher risk decreases it.

4.2 Action Bias Model (AB):

Behavioral economics posits that real-world decisions often deviate from pure risk–return optimization.

A minimal AB utility specification captures these effects:

$$U_{AB} = \lambda_1 \cdot BL - \lambda_2 \cdot exp + \lambda_3 \cdot lag_action$$

where all ($\lambda_i > 0$) measure the relative strength of each driver.

- Bright Line (BL): A significant trigger (e.g. price near a round number) that spurs action.
- Experience (exp): The accumulation of experience dampens impulsive actions, promoting reflection.
- Lagged Action (lag_action): Inertia from prior behavior amplifies the current action tendency.

This formulation illustrates the distortion introduced by action bias in decision-making: individuals tend to favor taking action driven by perceived subjective value, even in the absence of rational evaluation. Psychological triggers, such as the bright line threshold or behavioral inertia from previous actions, further reinforce this tendency, while accumulated experience may, to some extent, attenuate the bias.

Having completed the theoretical modeling, it is then necessary to test the proposed relationships and mechanisms through empirical methods. Therefore, the next section describes the selection of variables, the design of the regression model, details of the model set-up, and the empirical strategy used to analyse the determinants of behavioral biases and their behavioral differences from traditional utility theory.

5. REGRESSION

5.1 Data

Data source

The data set used in this analysis is extracted from Investing.com and focuses on the Shanghai Stock Exchange Composite Index ([SSEC](#)), which reflects the overall performance of all stocks listed on the Shanghai Stock Exchange and is a good indicator of general market sentiment or behavioral patterns. The data consists of daily trading information over 10 years from 29 December 2014 to 27 December 2024, totaling approximately 2400 trading days. Detailed price information is recorded for each observation, including opening prices, intraday highs and lows, closing prices, and trading volume.

Variable	N	Mean	Std. Dev.	Min	Max	Median
Closing Price	2432	3196.495	340.9876	2464.36	5166.35	3170.7
Opening Price	2432	3193.581	341.2040	2446.02	5174.42	3169.76
Highest Price	2432	3217.048	346.8974	2488.48	5178.19	3190.825
Lowest Price	2432	3170.427	332.0827	2440.91	5103.40	3151.725
Trade Volume	2432	28.3416	13.4102	7.06	131.35	26.42
Rise or Fall in Price	2432	0.0001	0.0131	-0.0849	0.0806	0.0005

Table 5.1: Descriptive Statistics of Original Market Data

Data process

1. Dependent Variable

- **action**: defined as a dummy variable representing the individual's action status, 1 if action is taken. This is calculated by comparing the trade volume today and three days before (most salient due to the PACF test), and is set to be 1 when an explicit increase in trade volume.

2. Important Independent Variables

- **BL (Psychological Threshold)**: defined as the psychological threshold as a

dummy variable reflecting the investor's propensity to trade when the price is close to a certain integer level or a significant psychological threshold.

- **exp (experience variable)**: defined as the cumulative level of market participation, which stretches from 1 to 10. Reflects the degree of investor experience by accumulating past investment behavior.

- **lag_action (lagged behavior)**: defined as the lagged value of the previous day's action variable, which is used to assess the persistence or inertia of investors' decisions.

- **lag_profit (marginal return proxy)**: defined as the difference between the today's closing price and the day before, representing the most recent realized return. It serves as a proxy for marginal utility in traditional utility theory, capturing the rational response of the investor to immediate past profitability.

Note: In this study, although the canonical utility specification of mean variation $U = \mathbb{E}[R] - \frac{1}{2}A\sigma^2$ provides a complete theoretical account of risk–return trade-offs, its key components, investor-specific risk aversion A and the full return distribution, are not directly observable in large-scale daily market data. To maintain both theoretical relevance and empirical feasibility, we therefore use the actual return of the prior period (**lag_profit**) as a proxy for marginal profit. Lagged profit can be calculated directly from publicly available price series, aligns intuitively with investors' perceptions of recent gains or losses, and has been widely used in the literature to test profit-driven trading behavior. By adopting **lag_profit**, we balance fidelity to the utility framework with data accessibility and model identifiability, ensuring that our Probit estimates credibly capture the influence of marginal returns on trading decisions.

3. Other Control Variables

- **lag_closing (lagged closing price)**: centered on the daily closing price variable (subtracted from the mean) and takes the lagged value of the previous trading day.

- **lag_highest (lagged highest price)**: the daily highest price variable is centered and takes the lagged value from the previous trading day.

- **lag_lowest**: centred on the daily low price variable with a lag of the previous

trading day.

- **lag3_log_tv (volume third-order lag term)**: due to the right-skewed distribution of the raw volume data, a natural logarithmic treatment is performed and is lagged by three periods to capture the continuation effect of past trading activity.

- **loss (loss aversion variable)**: defined as a dummy variable, assigned a value of 1 when the market return is negative (i.e, a loss occurs) and 0 otherwise, to examine the behavioral performance of the investor in a loss state.

Through the above data processing methods, this study constructs a variable system that meets the theoretical expectations and is more reasonable in its statistical nature. This system provides a solid data foundation for the subsequent regression analyzes.

Data test

1. Multicollinearity Test

This study performed a multicollinearity test on the independent variables included in the model to ensure the robustness of the regression results. Multicollinearity refers to strong linear relationships between independent variables in a regression model, which could result in inaccurate coefficient estimates, increased standard errors, and reduced reliability of regression outcomes.

This research used the Variance Inflation Factor (VIF) to detect multicollinearity, typically considering a VIF value greater than 10 indicative of severe multicollinearity. According to the outputs generated by Stata software, the VIF values for the critical explanatory variables used in this study — BL, exp, loss, and lag_action — were 1.00, 1.07, 1.00, and 1.07, respectively, with an average VIF of 1.04 (detailed test result see Appendix .6a), significantly below the threshold of 10. Hence, it can be concluded that there is no significant multicollinearity issue among the variables included in the model.

Therefore, multicollinearity does not affect the regression outcomes of this study, indicating that the independent variables selected for the model are appropriate.

2. Residual Autocorrelation Test

The residual autocorrelation test aims to verify the presence of serial correlation

among the residuals of the model. Residual autocorrelation can cause an underestimation of standard errors, leading to inaccurate significance tests for variables.

This study used the Durbin-Watson statistic (DW test) to detect residual autocorrelation. The regression results yielded a Durbin-Watson statistic of 1.9575, significantly close to 2. Typically, a Durbin-Watson statistic value between 1.5 and 2.5 indicates that there are no severe autocorrelation problems. Therefore, this model does not exhibit apparent residual autocorrelation.

In summary, the regression model developed in this study does not show significant serial correlation in its residuals, confirming that the estimation results are highly reliable.

5.2 Regression Design

To study the determinants of investor action and the extent of their influence on the probability of action, this study employs a Probit regression model, which is particularly well suited for modeling binary outcome variables. The binary dependent variable in this study, action, takes the value of 1 when an investor is observed to act (e.g., engage in a trade) and 0 otherwise. Traditional linear regression models (such as OLS) are inappropriate in this case, as they assume a continuous dependent variable and may predict probabilities outside the $[0,1]$ range. In contrast, the Probit model constrains the predicted values within this interval using the cumulative distribution function (CDF) of the standard normal distribution, making it ideal for capturing non-linear relationships in binary choice contexts.

Model Specification

The Probit model used in this study aims to estimate the probability that an investor will take an action on a given trading day, based on explanatory variables related to behavioral and market behavior. The regression function is formally expressed as:

$$\begin{aligned}
\text{action} = & \alpha + \beta_1 \text{BL} + \beta_2 \text{exp} + \beta_3 \text{lag_action} \\
& + \beta_4 \text{lag_highest} + \beta_5 \text{lag_lowest} + \beta_6 \text{opening} + \beta_7 \text{lag_closing} \\
& + \beta_8 \text{loss} + \beta_9 \text{lag3_log_tv}
\end{aligned}$$

Exogeneity and Identification Assumptions

To ensure unbiasedness and consistency of the estimated coefficients, it is assumed that all explanatory variables are exogenous, meaning that they are not correlated with the error term. This assumption is plausible in the current setting, as the explanatory variables are either predetermined (e.g., lagged values of market variables) or represent independent investor characteristics (such as experience).

- For example, lagged price variables and lagged actions occur before the observed action decision, satisfying the temporal precedence required for weak exogeneity.
- Additionally, variables such as BL and loss are constructed based on objective market movements rather than subjective perceptions, reducing the likelihood of endogeneity due to simultaneity or omitted variable bias.

Robustness checks, such as multicollinearity diagnostics and residual autocorrelation tests, further support the validity of the model structure and help ensure that coefficient estimates are not biased due to functional misspecification or intervariable dependencies.

Conclusion of Regression (Model) Design

By employing a Probit model with clearly defined behavioral and market control variables, this study builds a theoretically grounded and empirically rigorous framework for analyzing the probability of investor action. The model enables a direct comparison between predictions derived from behavioral economics (bright line, experience, lagged action) and traditional utility theory (marginal profit). This methodology provides a reliable basis for identifying which theoretical framework better explains observed decision-making behavior in financial markets.

5.3 Regression Result

Result Table

	AB				TU		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
action							
BL	0.174** (0.0874)	0.162* (0.0888)	0.202** (0.0916)	0.213** (0.0930)			
exp	-0.0515*** (0.0166)	-0.115*** (0.0177)	-0.0775*** (0.0186)	-0.0768*** (0.0190)			
lag_action		0.713*** (0.0545)	0.701*** (0.0568)	0.697*** (0.0567)			
lag_highest			-0.00157 (0.000999)	-0.0000527 (0.00108)		0.00346* (0.00182)	0.00324* (0.00182)
lag_lowest			-0.00395*** (0.00104)	-0.00227* (0.00118)		-0.00214 (0.00160)	-0.00191 (0.00161)
Opening			0.00249*** (0.000349)	0.00250*** (0.000463)		0.00207*** (0.000336)	0.00209*** (0.000337)
lag_closing			0.00367*** (0.00131)	0.000450 (0.00166)		-0.00271 (0.00307)	-0.00276 (0.00306)
lag3_log_tv			-0.841*** (0.0781)	-0.838*** (0.0760)		-0.937*** (0.0750)	-0.928*** (0.0750)
loss				-0.226*** (0.0672)			-0.210*** (0.0696)
lag_profit					0.00346*** (0.000661)	0.00424** (0.00172)	0.00243 (0.00183)
Constant	0.0511 (0.116)	0.0228 (0.118)	-5.416*** (1.126)	-5.360*** (1.490)	-0.0499* (0.0256)	-3.616*** (1.080)	-3.601*** (1.082)
Observations	2422	2422	2422	2422	2431	2429	2429

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

From the above table of regression results, it is clear that the factors of BL, exp and lag_action significantly influence the decision-making behavior of individuals in the context of Action Bias (AB). Among them, the direction of the BL coefficient is significantly positive, taking values between 0.162 and 0.213, and statistically significant ($p < 0.05$). This indicates that the closer an individual is to their psychologically set profit-and-loss boundary (Bright Line), the more inclined they are to take action, thus validating

psychologically driven decision-making bias. Experience (exp), on the other hand, is significantly negative, taking values between -0.0515 and -0.115 ($p < 0.01$), suggesting that the more experienced the individual, the more they tend to avoid blind action and prefer rational decision making. In addition, the effect of lag_action on current decision making is extremely significant, with the coefficient taking a value of about 0.7 or so ($p < 0.01$), fully reflecting the strong inertia and continuity characteristics in individual decision making.

In contrast, in the Traditional Utility (TU) context, individual behavior is mainly driven by marginal profit variables as lag_profit. When only the lag_profit variable is included in the model, its coefficient is 0.00424 and is significantly positive ($p < 0.05$). However, when the loss (loss) variable is included¹, the significance of lag_profit decreases rapidly (the coefficient decreases to 0.00243, which becomes statistically insignificant), reflecting that the effect of earnings is unstable under the traditional utility framework and is susceptible to the interference of psychological factors in the loss context. Meanwhile, other market variables such as opening price (opening) and highest price (lag_highest) have significant and stable impacts in the TU model, but the fluctuation of the coefficients is significantly smaller than that of the behavioral bias scenario, suggesting that the degree of fluctuation of individual behaviors in the TU model is more moderate and stable.

Next, we will explore each variable's marginal effect and predicted probability further to analyze the difference between the two models in terms of individual behavioral prediction ability and reveal the actual influence mechanism of behavioral economics factors on the decision-making process.

Marginal Effect

Figure 5.1 visually records the probabilistic impact of the action under various influencing factors. It is also known as the marginal impact chart. The first three BL, exp, and lag_action influence the action bias environment. Lag_profit influences the traditional utility, and loss is added to the discussion of marginal effects as the determining

¹Note: By explicitly controlling for the loss condition, the model now better isolates this psychological bias, demonstrating that the previously observed significance of lag_profit was, in part, capturing this behavioral response to losses. Without controlling for loss, the effects attributed to marginal profits were likely overstated, as the loss variable captures specific behavioral responses distinct from continuous profit levels.

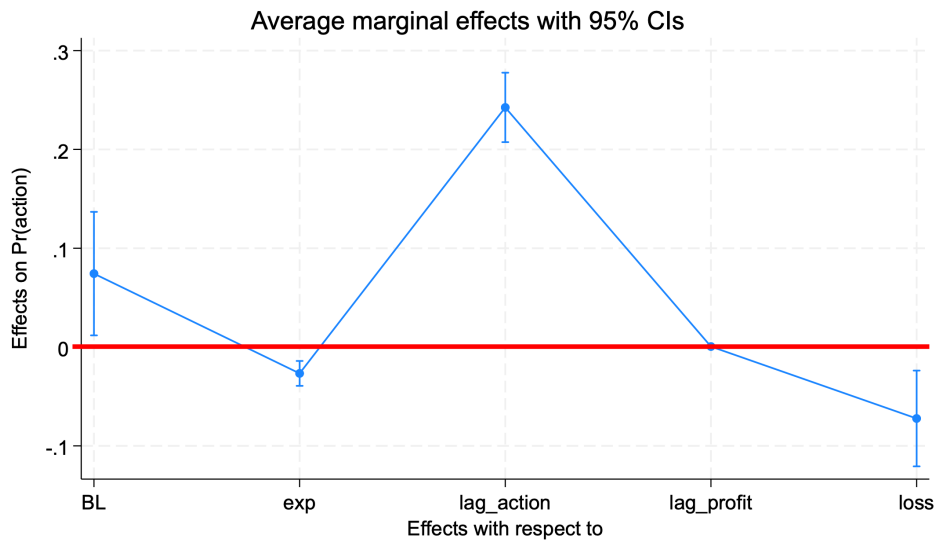


Figure 5.1: marginsplot

value of loss aversion.

First of all, we can observe the three influences of action bias, BL is a positive influence with a degree of 7.43%, which is significant at $p = 0.050$, although this degree is not large, but it is a good proof that people are more inclined to manipulate the capacity of the position when the yield fluctuation is close to the vicinity of people's psychological thresholds. Exp brings a negative influence with a degree of -2.67%, which is very significant ($p = 0.000$), again, not a large value, but illustrates that people tend to manipulate equity allocation less often when they gain more experience. Lag_action is the most influential component, with a degree of 24.3% and a very significant p-value ($p = 0.000$), implying that the probability of an individual taking action as a result of past actions increases by an average of 24.3 percentage points. This is a strong influence and may be critical in behavioral decision-making.

In sharp contrast, traditional utility theory posits that rational decisions are predominantly driven by marginal profits. Using lagged profit (lag_profit) as a proxy variable, empirical results reveal a marginal effect of merely 0.05%, which is not statistically significant ($p=0.418$). Further examination of the margins plot indicates that the effect associated with lag_profit is virtually at the zero line ($y = 0$). This underscores the negligible impact of marginal profit on individual decision-making behaviors, sharply contrasting the significant effects observed for psychological thresholds (BL), cumulative

experience (exp), and behavioral inertia (lag_action). In other words, according to predictions from traditional utility theory, individuals should significantly adjust their behavior in response to increases in profit. However, our data explicitly show that profit factors do not produce the expected influence.

Notice: the variable loss measures the degree of loss aversion; in this place, we use it only for marginal effect visualization, but not as a main variable for the study. The introduction of loss aversion is also used to explain its inverse effect on, for example, BL and lag_action, reducing the degree of its superimposed effect, and we can see that the marginal effect of loss is -7.23%, which is significant at $p = 5\%$, and is indeed aligned with our hypothesis that people will reduce the frequency of action due to loss aversion, although the result may be to hold losing stocks for long periods.

Building upon the marginal effect results, which quantify the average influence of each explanatory variable on the likelihood of investor action, we now turn to predicted probabilities to gain a more intuitive and context-specific understanding of how different behavioral conditions shape decision outcomes. Although marginal effects provide an overall estimate of sensitivity, predicted probabilities allow us to visualize and interpret how the probability of taking action evolves under varying levels or states of key variables such as BL, experience, and lagged behavior. This enables a more concrete behavioral interpretation of the results of the Probit model, highlighting not only whether a factor matters but also how it manifests in practical decision-making scenarios.

Predicted Probability (AB)

The regression results demonstrate the significant differences between action bias and traditional utility theory in explaining individual decision-making behaviors, particularly with respect to the degree to which various factors influence individual decisions.

Figure 5.2 illustrates the predicted probability of taking action as a function of the Bright Line (BL) variable, a psychological threshold indicator. The blue line on the graph represents the predicted probability $\Pr(\text{action})$, while the red line displays the regression-fit values. As BL increases from 0 to 1, indicating that an individual is closer to a psychological price threshold, the probability of taking action increases sharply from approximately 0.25 to over 0.80. This notable upward trend indicates a strong

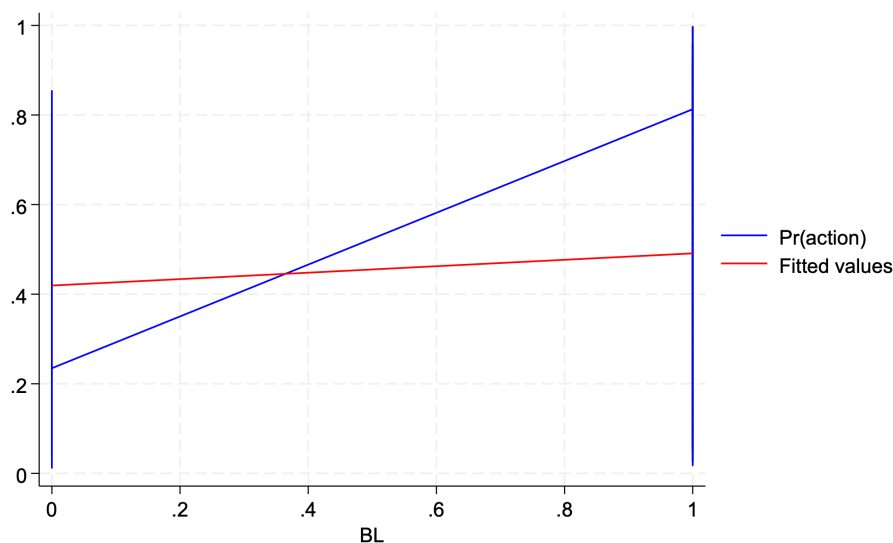


Figure 5.2: BL_pr

and statistically significant association between psychological price proximity and market activity. The fitted values exhibit a more moderate slope, which reinforces that even after adjusting for all covariates, the behavioral change induced by BL remains substantial. This visual outcome aligns with the marginal effects observed in the regression and confirms that crossing a perceived psychological threshold strongly influences investor behavior.

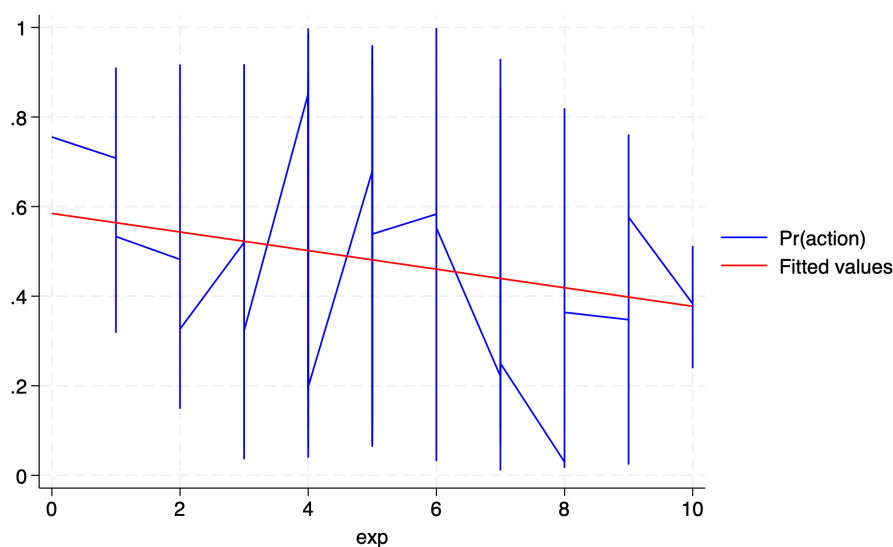


Figure 5.3: exp_pr

Figure 5.3 explores the predicted probability of taking action relative to the level of experience accumulated in investment (exp). The blue line captures the non-linear and

volatile behavior of actual predicted probabilities, whereas the red line shows a fitted decreasing trend. Although the plot fluctuates due to the nature of discrete experience data, a general downward slope is evident: as experience increases, the likelihood of taking action decreases (the fitted values go down from 0.60 to 0.40). This suggests that more experienced investors are less prone to reactive or potentially biased trading behaviors. This outcome substantiates the hypothesis that accumulated market participation tends to cultivate more rational and controlled decision-making patterns, diminishing impulsive responses in line with behavioral economic theory.

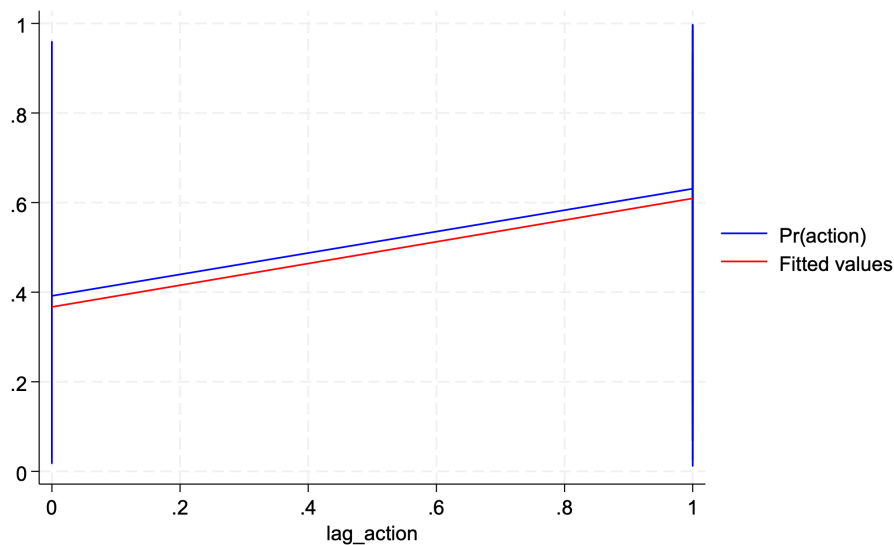


Figure 5.4: lag_action_pr

A notable discovery of this research is the substantial impact of lagged actions (lag_action) on current behavior, demonstrating significant behavioral continuity. The third plot depicts the predicted probability of action based on the action of the previous day (lag_action), a proxy for behavioral inertia. The blue line indicates that those who took action previously are significantly more likely to do so again, with the predicted probability steadily rising from approximately 0.40 to over 0.60 (the fitted values increase from 0.38 to 0.61). This strong relationship confirms the existence of behavioral persistence, commonly referred to as ‘action inertia’, where past behaviors serve as a key determinant of future decisions, independent of current market conditions. This insight provides empirical support for the theoretical framework that proposes that individuals are not only influenced by current signals but are also heavily anchored by their previous choices.

Predicted Probability (TU)

In the final regression model representing the traditional utility (TU) framework, `lag_profit`, which is a marginal utility proxy variable, is not statistically significant. Therefore, we did not perform any further individual-level predictive visualizations. However, from the regression results, we can see that the coefficient of `lag_profit` is positive but extremely small, indicating that its effect on the probability of investor action is minimal. This result contrasts with the behavioral variables - `BL`, `exp` and `lag_action`.

Conclusion

These empirical findings robustly illustrate the limitations of traditional utility theory in explaining actual investment behaviors and highlight the explanatory advantages of behavioral economic factors. Traditional theory assumes that individuals strictly follow profit maximization principles, yet in reality, investor behaviors are substantially influenced by psychological and experiential factors. The psychological threshold effect indicates investors do not continuously pursue profit maximization but are influenced by psychological benchmarks, cumulative experience effects reflect investors gradually adopting more cautious behaviors based on past experiences, and behavioral inertia reveals individuals are prone to repeating past decision patterns rather than dynamically adjusting decisions based on immediate profit fluctuations. These factors comprehensively illustrate the complexity of human decision-making behaviors, underscoring the need to incorporate behavioral economic perspectives in the analysis of real-world financial market behaviors.

In conclusion, relying solely on traditional economic utility frameworks is insufficient to fully describe and predict individual behaviors. In contrast, behavioral economics integrates psychological factors, cumulative experience, and behavioral inertia, offering a more precise and profound understanding. This integrated perspective is more applicable and guiding in practical financial decision-making analysis and policy making.

To connect our research findings with real-world relevance, it is important to examine how the identified behavioral patterns manifest in actual decision-making contexts. Therefore, the next section presents with real-world examples, illustrating how action bias influence behaviors within practical, everyday environments.

6. REAL WORLD APPLICATION

The practical implications of action bias in financial markets are not merely theoretical. Numerous real-world cases have demonstrated how psychological tendencies such as bright-line triggers (BL), experiential judgment (exp), and behavioral inertia (lag_action) can lead to significant inefficiencies in investor behavior, particularly during periods of market turbulence. Two notable cases from recent history, the COVID-19 pandemic and the 2008 global financial crisis, offer compelling evidence of action bias in large-scale investor responses.

During the COVID-19 pandemic, market participants showed strong signs of action bias in the form of panic sales. This widespread behavior was not primarily driven by rational reassessment of expected returns, but rather triggered by psychological thresholds, which we model as bright-line (BL) effects. For example, significant sell-offs occurred when stock prices approached psychologically salient round numbers, as many investors interpreted such levels as cues to act, even in the absence of updated fundamental information (Hong et al., 2021)[10]. The strong reaction at these thresholds reflects the influence of mental accounting and blending effects, which align with the determinant of BL of our model.

Moreover, accumulated experience (exp) appeared to moderate this action bias. Experienced investors showed more resilience, showing fewer impulsive actions compared to less experienced participants. This supports our regression findings, where experience was significantly negatively associated with the likelihood of taking action.

Another important factor is behavioral inertia (lag_action)—the tendency to repeat past actions. During both crises, many investors continued their selling or buying behavior merely because they had done so in previous sessions, reflecting inertia rather than real-time analysis. This pattern was particularly visible in the automated trading and retail investor segments, where algorithmic or emotional triggers often reinforced prior decisions (Barber & Odean, 2013)[1]. Our model captures this continuation through a strong and statistically significant coefficient on lag_action, further confirming its practical validity.

In sum, real-world observations validate our theoretical model of action bias centered on BL, exp, and lag_action. These behavioral determinants drive excessive and often

suboptimal activity in financial markets, especially during high-stress events. Understanding these dynamics is essential not only for academic modeling but also for practical investor education and policy design aimed at promoting more stable and rational market behavior.

7. CONCLUSION

Summary

This study delineates the distinct differences between action bias, as defined by behavioral economic theories, and the traditional utility theory concerning their explanatory power in individual decision-making behaviors. Empirical findings indicate that psychological thresholds, represented by the bright line (BL), significantly impact the probability of action, exhibiting a marginal effect of approximately 7.43% ($p=0.020$). Furthermore, investor experience (exp) is negatively correlated with the probability of action by about 2.67% ($p = 0.000$), which means that experienced individuals are less susceptible to behavioral biases. A crucial new finding in this investigation is the significant role of behavioral inertia, measured by lagged actions (lag_action), which dramatically influences current decisions with a marginal effect of approximately 24.25% ($p = 0.000$). These behavioral factors starkly contrast with the predictions from traditional utility theory, which posits that rational decisions are primarily driven by marginal profit. Our analysis using lagged profit (lag_profit) shows negligible and statistically insignificant effects, reinforcing the inadequacy of traditional profit maximization frameworks in explaining actual observed behaviors. In general, behavioral economic factors such as psychological thresholds, experience, and inertia offer substantially greater explanatory power and practical relevance than traditional utility models.

Limitation

Despite the robust findings, this study also has several limitations. First, the analysis is based on daily data from the Shanghai Stock Exchange Composite Index (SSEC) that span a decade, which may restrict the generalizability of the conclusions to other market contexts or time horizons. Secondly, the variables chosen to represent psychological thresholds and experience, while theoretically justified, could still be subject to measurement errors or conceptual simplifications. Future studies could incorporate more refined proxy models or direct psychological measurements to improve precision. Third, this research employed a binary Probit model, suitable for capturing the dichotomous nature of trading decisions but potentially overlooking nuanced continuous variations in investor behaviors. Alternative

econometric models, including panel data analysis or structural equation modeling, could provide deeper insights into investor behavior dynamics. Lastly, this study focuses primarily on individual-level decision-making factors and excludes macroeconomic and institutional factors that could also substantially influence investor behavior. Future research should consider integrating broader economic contexts and regulatory influences to develop a more comprehensive behavioral-economic model.

Further research suggestions

Several avenues for future research could be explored to expand and enhance the findings of this study. First, from the perspective of data collection, the acquisition of individual-level data would significantly improve the accuracy and robustness of regression analyzes. By capturing detailed individual behavioral patterns rather than aggregated data, researchers can more precisely identify the nuances of decision-making behaviors. In addition, employing carefully designed questionnaires or structured interviews can help capture the true psychological motivations, intentions, and thresholds of individuals, providing more deep insight into the underlying cognitive biases.

Secondly, with regard to regression methodologies, future research could benefit from advanced econometric techniques specifically tailored to individual-level panel data, such as mixed effects models or hierarchical linear modeling. These models would allow for more precise accounting of heterogeneity across individuals, controlling for both fixed and random effects. In addition, advanced robustness checks, including instrumental variable regressions or propensity score matching, could further verify causality and mitigate potential endogeneity issues in observed relationships.

Thirdly, on the application front, it would be worthwhile to extend the scope of the study beyond financial stock markets to other decision-making contexts. Possible fields of application could include real estate markets, consumer purchase behaviors, healthcare decisions, and public policy choices, where understanding the impact of cognitive biases such as action bias could significantly improve outcomes. Importantly, the manifestation and intensity of action bias can vary between these domains, depending on the stakes of the decision, emotional involvement, and the availability of information. Recognizing these contextual differences would allow for more tailored interventions and better informed

behavioral policies.

Lastly, future research should aim to more accurately delineate the drivers of action bias and quantify their effects with greater precision. By conducting more controlled and nuanced experiments, researchers can better understand the mechanisms behind action bias and offer practical evidence-based recommendations to mitigate its adverse effects. Ultimately, providing the general public with scientifically validated strategies to reduce cognitive biases will lead to more informed, rational, and beneficial decision-making practices.

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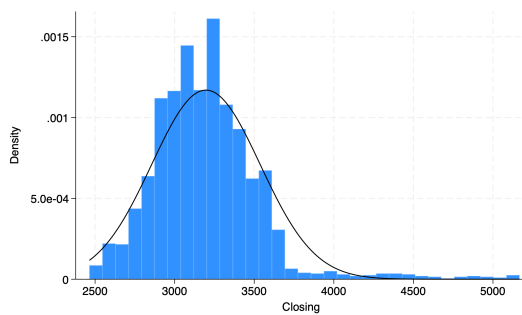
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APPENDIX

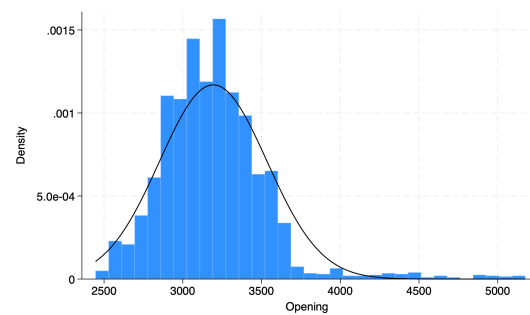
A. Original Data Display

Observations:	2,432			
Variables:	7			
Variable name	Storage type	Display format	Value label	Variable label
date	str10	%10s		Date
closing	float	%9.0g		Closing
opening	float	%9.0g		Opening
highest	float	%9.0g		Highest
lowest	float	%9.0g		Lowest
tradevolume	float	%9.0g		TradeVolume
riseorfallinprice	float	%9.0g		rise or fall in price

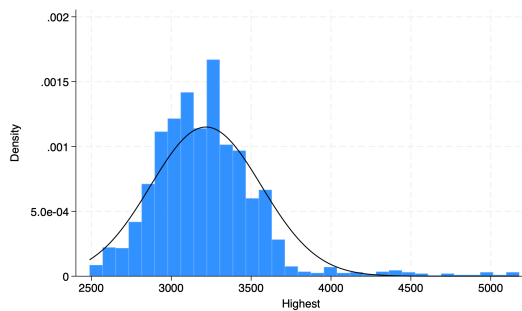
Figure .1: Original Data Describe



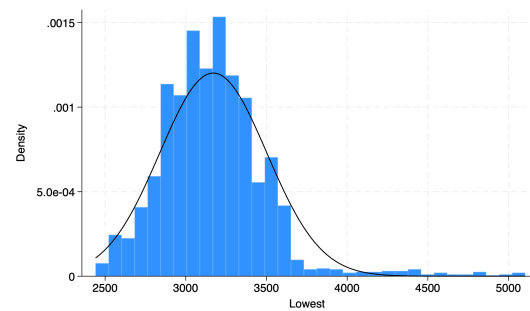
(a) Original Data: Closing



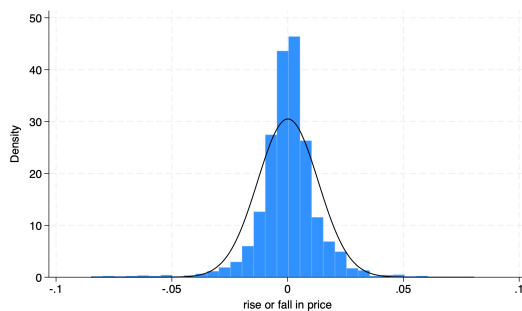
(b) Original Data: Opening



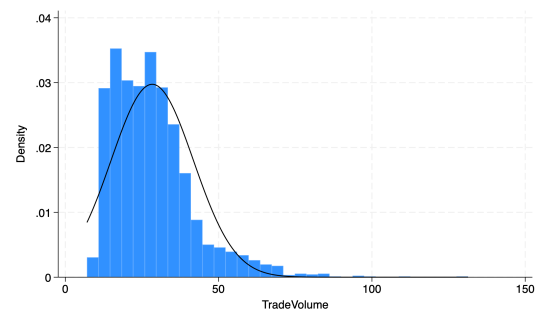
(c) Original Data: Highest



(d) Original Data: Lowest



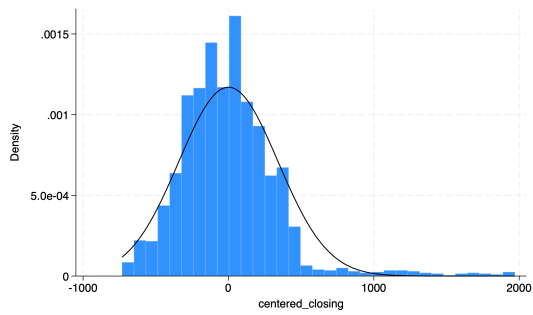
(e) Original Data: Rise or Fall in Price



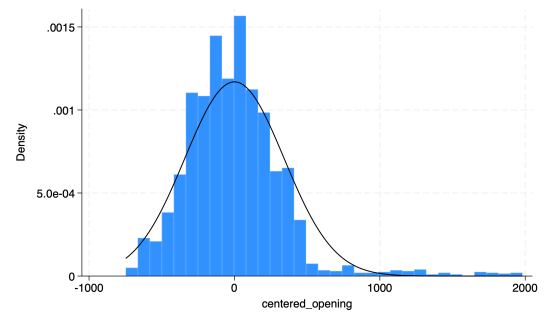
(f) Original Data: Trade Volume

Figure .2: Original Data Normal Diagrams

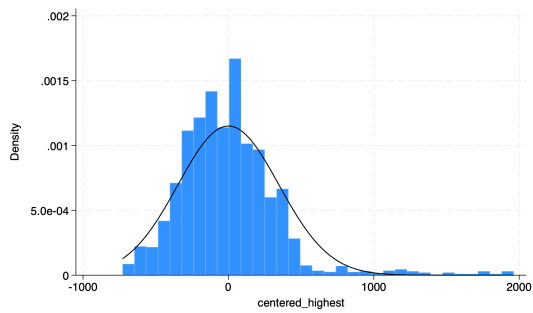
B. Centralized Processed Data Display



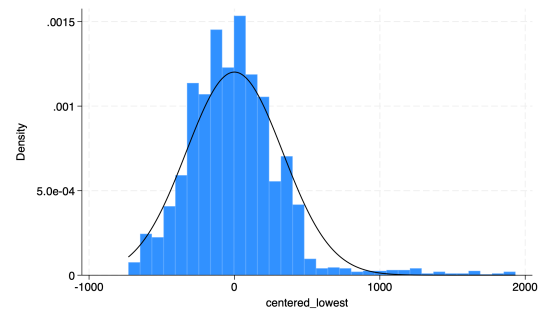
(a) Processed Data: Centralized Closing



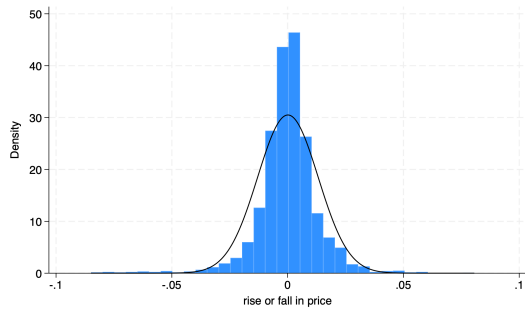
(b) Processed Data: Centralized Opening



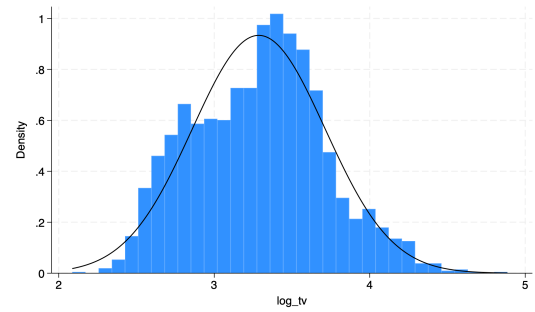
(c) Processed Data: Centralized Highest



(d) Processed Data: Centralized Lowest



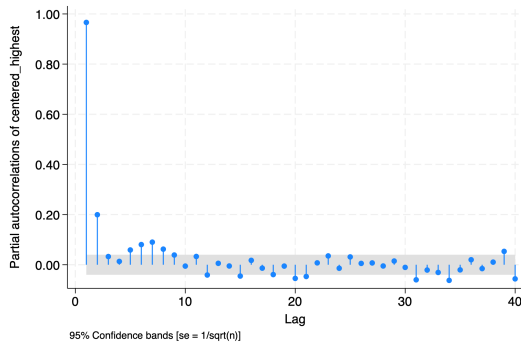
(e) Processed Data: Rise or Fall in Price



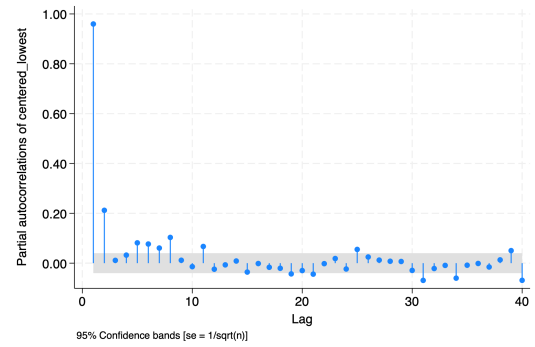
(f) Processed Data: log trade volume

Figure .3: Centralized Processed Data Normal Diagrams

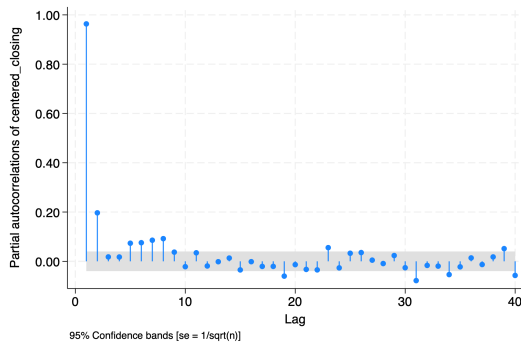
C. PACF Set Order Display



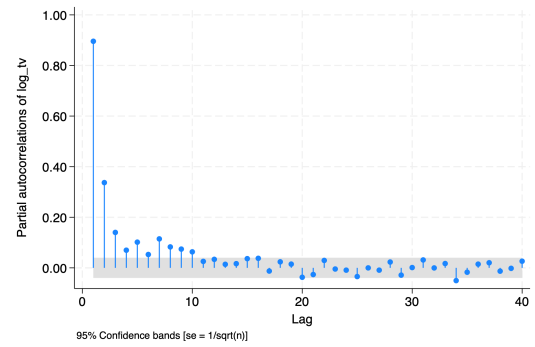
(a) Centralized Highest Lag Order: 1



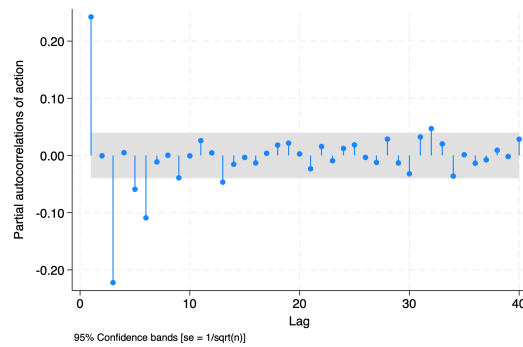
(b) Centralized Lowest Lag Order: 1



(c) Centralized Closing Lag Order: 1



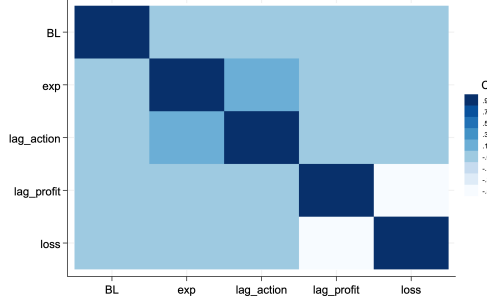
(d) Centralized Highest Lag Order: 3



(e) Action Lag Order: 1

Figure .4: PACF Set Order

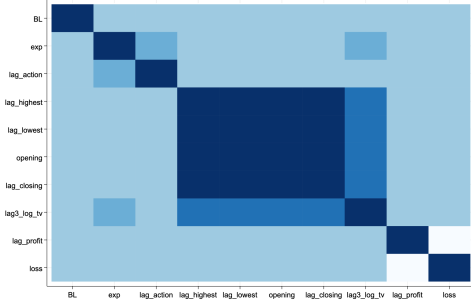
D. Correlation Test Display



(a) Correlation of Core Variables

	BL	exp	lag_ac~n	lag_pr~t	loss
BL	1.0000				
exp	-0.0187	1.0000			
lag_action	0.0108	0.2551	1.0000		
lag_profit	-0.0117	0.0162	0.0376	1.0000	
loss	0.0263	-0.0098	-0.0526	-0.6467	1.0000

(b) Correlation of Core Variables Table

(c) Correlation of All Variables ²

	BL	exp	lag_ac~n	lag_hi~t	lag_lo~t	opening	lag_cl~g		
BL	1.0000								
exp	-0.0187	1.0000							
	0.3564								
lag_action	0.0108	0.2551	1.0000						
	0.5951	0.0000							
lag_highest	0.0268	0.0049	0.0542	1.0000					
	0.1073	0.0000	0.0076						
lag_lowest	0.0191	0.0013	0.0437	0.9943	1.0000				
	0.3450	0.0001	0.0312	0.0000					
opening	0.0300	0.0766	0.0339	0.9644	0.9620	1.0000			
	0.1290	0.0002	0.0944	0.0000	0.0000				
lag_closing	0.0228	0.0042	0.0512	0.9965	0.9964	0.9677	1.0000		
	0.2612	0.0000	0.0115	0.0000	0.0000	0.0000		lag3_l~v	lag_pr~t
									loss
lag3_log_tv	0.0490	0.2217	-0.0204	0.5478	0.5313	0.5309	0.5308	1.0000	
	0.0141	0.0000	0.3149	0.0000	0.0000	0.0000	0.0000		
lag_profit	-0.0117	0.0162	0.0376	-0.0127	-0.0035	0.0062	0.0514	-0.0013	1.0000
	0.5034	0.4268	0.0641	0.5304	0.0650	0.0030	0.0113	0.9500	
loss	0.0263	-0.0098	-0.0526	-0.0543	-0.0539	-0.0920	-0.0927	-0.0110	-0.6467
	0.1955	0.6303	0.0095	0.0074	0.0078	0.0000	0.0000	0.5863	0.0000

(d) Correlation of All Variables Table

Figure .5: Overview of Four Comparative Results

²Note: The correlation between lag_highest, lag_lowest, opening, lag_closing and each other is relatively large, and there may be a problem of covariance, but the regression used in this paper did not find that this problem exists. These variables are used mainly as control variables, not core variables, so this paper does not deal with these variables, but considering the rigor, this paper suggests that future research can be done to deal with these variables appropriately in order to reduce the correlation.

G. Dofile Display

```
* Use the daily data
import delimited "/Users/sketchysaturn92/Desktop/SSEC_daily.csv",
    clear
save "/Users/sketchysaturn92/Desktop/SSEC_daily.dta", replace
use "/Users/sketchysaturn92/Desktop/SSEC_daily.dta", clear

* show the data
describe
summarize, detail

* sort the time
sort date
gen time = _n
tsset time

* test if normal
histogram closing, normal
histogram opening, normal
histogram highest, normal
histogram lowest, normal
histogram tradevolume, normal

* sort the skewed data
gen log_tv = log(tradevolume)
histogram log_tv, normal
summarize closing
gen centered_closing = closing - r(mean)
histogram centered_closing, normal
summarize opening
gen centered_opening = opening - r(mean)
histogram centered_opening, normal
summarize highest
```

```
gen centered_highest = highest - r(mean)
histogram centered_highest, normal
summarize lowest
gen centered_lowest = lowest - r(mean)
histogram centered_lowest, normal

* PACF set order
pac centered_closing,
pac centered_highest,
pac centered_lowest,
pac log_tv,

* generate action
gen action = 1
gen lag3_log_tv = L3.log_tv
gen compare = log_tv - lag3_log_tv
replace action = 0 if compare <= 0

* generate lag closing
gen lag_closing = centered_closing[_n-1]
gen lag_opening = centered_opening[_n-1]

* generate lag highest and lowest
gen lag_highest = centered_highest[_n-1]
gen lag_lowest = centered_lowest[_n-1]

* generate Bright line
gen BL = 0
gen diffoc = 0
replace diffoc = (closing-opening)/opening
replace diffoc = diffoc*100
egen diffoc_m = mean(diffoc)
replace BL = 1 if abs(diffoc) >= abs(diffoc_m)
```

```
* generate experience
gen exp = L1.action + L2.action + L3.action + L4.action + L5.
        action + L6.action + L7.action + L8.action + L9.action + L10.
        action

* generate loss
gen loss = 0
gen lagco_diff = lag_closing-lag_opening
replace loss = 1 if lagco_diff < 0

* generate lag_action
gen lag_action = action[_n-1]

* correlation Test
//important variable test
pwcorr BL exp lag_action lag_profit loss, sig
matrix C = r(C)
heatplot C, color(Blues)
//all variable test
pwcorr BL exp lag_action lag_highest lag_lowest opening
        lag_closing lag3_log_tv lag_profit loss, sig
matrix C = r(C)
heatplot C, color(Blues)

* data test
//Multicollinearity Test
reg action BL exp loss lag_action
estat vif
//Residual Autocorrelation Test
reg action BL exp loss lag_highest lag_lowest opening lag_closing
        lag_action log_tv
dwstat

* AB probit regression
```

```
probit action BL exp
est store model1
probit action BL exp lag_action
est store model2
probit action BL exp lag_action lag_highest lag_lowest opening
    lag_closing lag3_log_tv
est store model3
probit action BL exp lag_action lag_highest lag_lowest opening
    lag_closing loss lag3_log_tv, vce(robust)
est store model4
margins, dydx(BL)
margins, dydx(exp)
margins, dydx(loss)
margins, dydx(lag_action)

* AB graph
predict phat, pr
twoway (line phat BL, sort lcolor(blue) lwidth(medium))(lfit phat
    BL, lcolor(red))
twoway (line phat lag_action, sort lcolor(blue) lwidth(medium))(
    lfit phat lag_action, lcolor(red))
twoway (line phat exp, sort lcolor(blue) lwidth(medium))(lfit
    phat exp, lcolor(red))

* TU variable create: lag_profit
gen lag_opening1 = opening[_n-1]
gen lag_closing1 = closing[_n-1]
gen lag_profit = lag_closing1 - lag_opening1

* TU probit regression
probit action lag_profit
est store model5
probit action lag_profit lag_highest lag_lowest opening
    lag_closing lag3_log_tv
```



```
est store model6
probit action lag_profit lag_highest lag_lowest opening
      lag_closing loss lag3_log_tv
est store model7
margins, dydx(lag_profit)

* TU graph
predict phat1, pr
twoway (line phat1 lag_profit, sort lcolor(blue) lwidth(medium))(
      lfit phat1 lag_profit, lcolor(red))

* egression result
esttab model1 model2 model3 model4 model5 model6 model7, se star
      (* 0.1 ** 0.05 *** 0.01) label booktabs

* margins diagram
probit action BL exp lag_action lag_highest lag_lowest opening
      lag_closing lag3_log_tv lag_profit loss
margins, dydx(BL exp lag_action lag_profit loss)
marginsplot
```

Listing 1: Excerpt from Stata Do-File